

[MS-MWBE]:

Microsoft Web Browser Federated Sign-On Protocol Extensions

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1 Introduction

This specification extends the Microsoft Web Browser Federated Sign-On Protocol described in [\[MS-MWBF\]](#). It is assumed that the reader is familiar with its terms, concepts, and protocols.

The extensions defined in this specification enable **web browser requestors** that do not support scripting (to create POST messages) and enable passing **security identifiers (SIDs)** in Security Assertion Markup Language (SAML) 1.1 assertions. These extensions are referred to, respectively, as the Query String Response Transfer Protocol and the SAML 1.1 Assertion Extension.

The Microsoft Web Browser Federated Sign-On Protocol specifies the use of HTTP POST to transmit the **wsignin1.0** result. The use of HTTP POST requires web browser requestors to support scripting for automated form submittal, but web browser requestors do not always have scripting support. The Query String Response Transfer Protocol provides a method for using a series of HTTP GET messages instead of a single HTTP POST to transmit the result of a wsignin1.0 action. This eliminates the scripting requirement for the web browser requestor. That is, the extension increases the number of messages needed to perform a wsignin1.0 action to avoid the POST message.

The SAML 1.1 Assertion Extension is an extension of the Microsoft Web Browser Federated Sign-On Protocol that specifies a method for transmitting SIDs as elements in **SAML advice**.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

account: A **user** (including machine account), group, or alias object. Also a synonym for security principal or principal.

Active Directory: The Windows implementation of a general-purpose directory service, which uses **LDAP** as its primary access protocol. Active Directory stores information about a variety of objects in the network such as user accounts, computer accounts, groups, and all related credential information used by Kerberos [\[MS-KILE\]](#). Active Directory is either deployed as Active Directory Domain Services (AD DS) or Active Directory Lightweight Directory Services (AD LDS), which are both described in [\[MS-ADOD\]](#): Active Directory Protocols Overview.

aggregated result: The assembly of received parts transferred using the Query String Response Transfer Protocol. The **aggregated result** is assembled at a **relying party** and might not represent the complete result if all parts have not been received. Once complete, the **relying party** extracts a **RequestSecurityTokenResponse (RSTR)** from the **aggregated result**. For more information, see section 3.2.1.1.1.

base64 encoding: A binary-to-text encoding scheme whereby an arbitrary sequence of bytes is converted to a sequence of printable ASCII characters, as described in [\[RFC4648\]](#).

claim: A declaration made by an entity (for example, name, identity, key, group, privilege, and capability). For more information, see [\[WSFederation1.2\]](#).

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a domain controller (DC) and host a member list that identifies all members of the domain, as well as optionally hosting the **Active Directory** service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

forest: One or more **domains** that share a common schema and trust each other transitively. An organization can have multiple **forests**. A **forest** establishes the security and administrative

boundary for all the objects that reside within the **domains** that belong to the **forest**. In contrast, a **domain** establishes the administrative boundary for managing objects, such as users, groups, and computers. In addition, each **domain** has individual security policies and trust relationships with other **domains**.

global group: An **Active Directory** group that allows user objects from its own **domain** and **global groups** from its own **domain** as members. Also called domain global group. **Universal groups** can contain **global groups**. A group object *g* is a **global group** if and only if `GROUP_TYPE_ACCOUNT_GROUP` is present in *g*! `groupType`; see [MS-ADTS] section 2.2.12, "Group Type Flags". A **global group** that is also a security-enabled group is valid for inclusion within ACLs anywhere in the **forest**. If a **domain** is in mixed mode, then a **global group** in that **domain** that is also a security-enabled group allows only user object as members. See also domain local group, security-enabled group.

identity provider/security token service (IP/STS): An STS that might also be an identity provider (IP). This term is used as shorthand to see both identity that verifies token services and general token services that do not verify identity. Note that the "/" symbol implies an "or" relationship.

Lightweight Directory Access Protocol (LDAP): The primary access protocol for **Active Directory**. Lightweight Directory Access Protocol (LDAP) is an industry-standard protocol, established by the Internet Engineering Task Force (IETF), which allows users to query and update information in a directory service (DS), as described in [MS-ADTS]. The Lightweight Directory Access Protocol can be either version 2 [\[RFC1777\]](#) or version 3 [\[RFC3377\]](#).

little-endian: Multiple-byte values that are byte-ordered with the least significant byte stored in the memory location with the lowest address.

NetBIOS: A particular network transport that is part of the LAN Manager protocol suite. **NetBIOS** uses a broadcast communication style that was applicable to early segmented local area networks. A protocol family including name resolution, datagram, and connection services. For more information, see [\[RFC1001\]](#) and [\[RFC1002\]](#).

pending result: The transformed **RequestSecurityTokenResponse (RSTR)** that an **identity provider/security token service (IP/STS)** maintains for the duration of a Query String Response Transfer Protocol message series. Each message in the Query String Response Transfer Protocol transfers a portion of the **pending result** to the **relying party**, where the portions are assembled into the **aggregated result**. For more information, see section 3.1.1.1.1.

relative identifier (RID): The last item in the series of SubAuthority values in a **security identifier (SID)** [\[SID\]](#). It distinguishes one account or group from all other accounts and groups in the domain. No two accounts or groups in any domain share the same RID.

relying party (RP): A web application or service that consumes **security tokens** issued by a security token service (STS).

requestor IP/STS: An **IP/STS** in the same **security realms** as the **web browser requestor**. The **requestor IP/STS** has an existing relationship with the **user** that enables it to issue **security tokens** containing **user** information.

RequestSecurityTokenResponse (RSTR): An XML element used to return an issued **security token** and associated metadata. An **RSTR** element is the result of the **wsignin1.0** action in the Web Browser Federated Sign-On Protocol. For more information, see [\[MS-MWBF\]](#) section 2.2.4.1.

resource IP/STS: An **IP/STS** in the same **security realm** as the **web service (WS) resource**. The **resource IP/STS** has an existing relationship with the **WS resource** that enables it to issue **security tokens** that are trusted by the **WS resource**.

SAML advice: The advice element of a **SAML assertion**. The data in the advice element is advisory and can be ignored without affecting the validity of the assertion. See [\[SAMLCore\]](#) section 2.3.2.2. The SAML 1.1 Assertion Extension includes **security identifiers (SIDs)** and related data in the **SAML advice** element.

SAML assertion: The Security Assertion Markup Language (SAML) 1.1 assertion is a standard XML format for representing a **security token**. For more information, see [\[SAMLCore\]](#) section 2.

security identifier (SID): An identifier for security principals that is used to identify an account or a group. Conceptually, the **SID** is composed of an account authority portion (typically a **domain**) and a smaller integer representing an identity relative to the account authority, termed the **relative identifier (RID)**. The **SID** format is specified in [\[MS-DTYP\]](#) section 2.4.2; a string representation of **SIDs** is specified in [\[MS-DTYP\]](#) section 2.4.2 and [\[MS-AZOD\]](#) section 1.1.1.2.

security realm or security domain: Represents a single unit of security administration or trust, for example, a Kerberos realm (for more information, see [\[RFC4120\]](#)) or a Windows Domain (for more information, see [\[MSFT-ADC\]](#)).

security token: A collection of one or more **claims**. Specifically in the case of mobile devices, a **security token** represents a previously authenticated user as defined in the Mobile Device Enrollment Protocol [\[MS-MDE\]](#).

subject: The entity to which the **claims** and other data in a **SAML assertion** apply. For more information, see [\[SAMLCore\]](#) section 1.3.1.

trusted forest: A forest that is trusted to make authentication statements for security principals in that forest. Assuming forest A trusts forest B, all domains belonging to forest A will trust all domains in forest B, subject to policy configuration.

universal group: An **Active Directory** group that allows user objects, **global groups**, and **universal groups** from anywhere in the **forest** as members. A group object g is a **universal group** if and only if GROUP_TYPE_UNIVERSAL_GROUP is present in g! groupType. A security-enabled universal group is valid for inclusion within ACLs anywhere in the **forest**. If a **domain** is in mixed mode, then a **universal group** cannot be created in that **domain**. See also domain local group, security-enabled group.

user: A person who employs a **web browser requestor** to access a **WS resource**.

user agent: An HTTP user agent, as specified in [\[RFC2616\]](#).

web browser requestor: An HTTP 1.1 web browser client that transmits protocol messages between an **IP/STS** and a **relying party**.

web service (WS) resource: A destination HTTP 1.1 web application or an HTTP 1.1 resource serviced by the application. In the context of this protocol, it refers to the application or manager of the resource that receives identity information and assertions issued by an **IP/STS** using this protocol. The **WS resource** is a **relying party** in the context of this protocol. For more information, see [\[WSFederation1.2\]](#).

wsignin1.0: A protocol message exchange defined in [\[WSFederation1.2\]](#) sections 2.1 and 3.1. The **wsignin1.0** request and response are the HTTP binding for the WS-Trust Issue action and response; as such, the WS-Trust **RSTR** element is used to return the issued **security token** in the **wsignin1.0** response ([\[WSTrust\]](#) section 3.2). For more information, see [\[MS-MWBF\]](#) section 2.2.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-ADA1] Microsoft Corporation, "[Active Directory Schema Attributes A-L](#)".

[MS-ADA3] Microsoft Corporation, "[Active Directory Schema Attributes N-Z](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-MWBF] Microsoft Corporation, "[Microsoft Web Browser Federated Sign-On Protocol](#)".

[RFC1950] Deutsch, P., and Gailly, J-L., "ZLIB Compressed Data Format Specification version 3.3", RFC 1950, May 1996, <http://www.ietf.org/rfc/rfc1950.txt>

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[RFC2616] Fielding, R., Gettys, J., Mogul, J., et al., "Hypertext Transfer Protocol -- HTTP/1.1", RFC 2616, June 1999, <https://www.rfc-editor.org/info/rfc2616>

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[SAMLCore] Maler, E., Mishra, P., Philpott, R., et al., "Assertions and Protocol for the OASIS Security Assertion Markup Language (SAML) V1.1", September 2003, <http://www.oasis-open.org/committees/download.php/3406/oasis-sstc-saml-core-1.1.pdf>

[WSFederation1.2] Kaler, C., McIntosh, M., "Web Services Federation Language (WS-Federation)", Version 1.2, May 2009, <http://docs.oasis-open.org/wsfed/federation/v1.2/os/ws-federation-1.2-spec-os.html>

[WSFederation] Kaler, C., Nadalin, A., Bajaj, S., et al., "Web Services Federation Language (WS-Federation)", Version 1.1, December 2006, <http://specs.xmlsoap.org/ws/2006/12/federation/ws-federation.pdf>

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[XML] World Wide Web Consortium, "Extensible Markup Language (XML) 1.0 (Fourth Edition)", W3C Recommendation 16 August 2006, edited in place 29 September 2006, <http://www.w3.org/TR/2006/REC-xml-20060816/>

1.2.2 Informative References

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[IANASCHEME] IANA, "Uniform Resource Identifier (URI) Schemes per RFC4395", November 2006, <http://www.iana.org/assignments/uri-schemes/uri-schemes.xhtml>

[MAXURL] Microsoft Corporation, "Maximum URL Length Is 2,083 Characters in Internet Explorer", September 2009, <https://support.microsoft.com/en-us/help/208427/maximum-url-length-is-2,083-characters-in-internet-explorer>

[MS-ADTS] Microsoft Corporation, "[Active Directory Technical Specification](#)".

[MS-PAC] Microsoft Corporation, "[Privilege Attribute Certificate Data Structure](#)".

[MS-WPO] Microsoft Corporation, "[Windows Protocols Overview](#)".

[SIDD] Microsoft Corporation, "How Security Identifiers Work", March 2003, <http://technet.microsoft.com/en-us/library/cc778824.aspx>

1.3 Overview

This document specifies the Query String Response Transfer Protocol and the SAML 1.1 Assertion Extension. These extensions are based on the federated sign-on protocol described in [\[MS-MWBF\]](#). The extensions specified in this document broaden the applicability of the protocol to simpler **web browser requestors** and a wider range of protected applications. The extensions in this specification do not change the services of authentication, identity federation, or single sign-on provided by [MS-MWBF].

1.3.1 Query String Response Transfer Protocol

The scripting capability for forms submittal is not specified as part of HTTP [\[RFC2616\]](#); consequently, not all **web browser requestor** implementations support forms submittal as recommended for **wsignin1.0** responses in [\[MS-MWBF\]](#) section 2.1. In addition, the **RequestSecurityTokenResponse (RSTR)** can be too large to be transferred in a single HTTP GET message. The Query String Response Transfer Protocol addresses these issues by eliminating the web browser requestor requirement for scripting support.

When using the Query String Response Transfer Protocol, the wsignin1.0 message exchange in [MS-MWBF] is replaced by a series of wsignin1.0 message exchanges. In [MS-MWBF], the RSTR is transmitted using a single HTTP POST message. The Query String Response Transfer Protocol transmits the RSTR in pieces in query string parameters using multiple HTTP GET messages. The **relying party** accumulates the pieces from the responses in the series to produce an **aggregated result**. When the relying party has accumulated all the pieces, it extracts the RSTR from the aggregated result.

1.3.2 SAML 1.1 Assertion Extension

The Microsoft Web Browser Federated Sign-On Protocol described in [\[MS-MWBF\]](#) does not specify a method for including **SIDs** in a **security token**. For applications requiring SIDs, **claims** are not sufficient for authorization.

The SAML 1.1 Assertion Extension provides a method for including SIDs in a **SAML assertion**. How an **identity provider/security token service (IP/STS)** obtains the security identifiers and how a **relying party** interprets them is implementation-specific.

1.4 Relationship to Other Protocols

The Web Browser Federated Sign-On Protocol and the extensions specified in this document use standard web protocols, XML (as specified in [\[XML\]](#)), WS-Federation Passive Requestor Profile (as specified in [\[WSFederation1.2\]](#) section 13), and SAML 1.1 (as specified in [\[SAMLCore\]](#)). The reader has to be familiar with the specifications listed in [\[MS-MWBF\]](#) section 1.4.

A **relying party** uses the Query String Response Transfer Protocol instead of the **wsignin1.0** messages from Web Browser Federated Sign-On Protocol to avoid the use of HTTP POST messages.

The SAML 1.1 Assertion Extension provides a facility for including **security identifiers** in **SAML assertions**. In order to understand the Windows behavior relating to this extension, the reader has to be familiar with the Active Directory Technical Specification [\[MS-ADTS\]](#), security concepts in [\[MS-WPO\]](#) section 9, and security identifiers [\[SID\]](#).

The Web Browser Federated Sign-On Protocol and the extensions specified in this document can be applicable where other web-based authentication protocols are used. For more information, see [\[MS-MWBF\]](#) section 1.4.

1.5 Prerequisites/Preconditions

The SAML 1.1 Assertion Extension requires that an **IP/STS** have a source of **security identifiers** and that the **relying party** have an authorization framework in which to interpret them. The exact methods by which the IP/STS obtains the SIDs, and the methods by which the relying party interprets them, are implementation-specific. [<1><2>](#)

1.6 Applicability Statement

The Query String Response Transfer Protocol is applicable where the Web Browser Federated Sign-On Protocol described in [\[MS-MWBF\]](#) is applicable. [<3>](#) The Query String Response Transfer Protocol widens the applicability of the Microsoft Web Browser Federated Sign-On Protocol to include **web browser requestors** that do not implement scripting or form submittal via scripting.

The SAML 1.1 Assertion Extension is applicable when the protected HTTP web application requires **SIDs** to perform authorization. [<4>](#)

1.7 Versioning and Capability Negotiation

The Web Browser Federated Sign-On Protocol as described in [\[MS-MWBF\]](#) section 1.7 defers all versioning and capability negotiation to [\[WSFederation\]](#), [\[WSFederation1.2\]](#), and [\[RFC2616\]](#).

When using the Query String Response Transfer Protocol, an **IP/STS** uses the presence of the *ttpindex* parameter (as specified in section [2.2.2.1.1](#)) in the request to determine whether the Query String Response Transfer Protocol is used for the response.

The SAML 1.1 Assertion Extension uses **SIDs**. SIDs have a revision mechanism. For more information, see [\[MS-DTYP\]](#), as specified in section 2.1.

1.8 Vendor-Extensible Fields

The extensions specified in this document make use of vendor-extensible fields that are specified as part of [\[MS-MWBF\]](#) section 1.8. Specifically, the Query String Response Transfer Protocol specifies new message parameters as allowed by [\[WSFederation1.2\]](#) section 13.2.1, and the SAML 1.1 Assertion Extension specifies new elements in the **SAML advice** as allowed by [\[SAMLCore\]](#) section 2.3.2.2.

The Query String Response Transfer Protocol does not introduce any vendor-extensible fields that are not present in [\[MS-MWBF\]](#) section 1.8.

The SAML 1.1 Assertion Extension introduces the [ClaimSource \(section 3.1.5.2.1.1\)](#) element whose value is a URI that can be extended by vendors. Uniqueness of URIs is scheme-dependent. For more information, see [\[IANAScheme\]](#) and [\[RFC4395\]](#).

1.9 Standards Assignments

There are no standards assignments beyond those for XML namespaces and standard ports specified in [\[MS-MWBF\]](#) section 1.9. [<5>](#)

2 Messages

The Query String Response Transfer Protocol and SAML 1.1 Assertion Extension extend the messages specified in [\[MS-MWBF\]](#) section 2 as described in this section.

2.1 Transport

No additional transport is required other than that provided for in [\[MS-MWBF\]](#) section 2.1.

2.1.1 Query String Response Transfer Protocol

In the Query String Response Transfer Protocol, all **wsignin1.0** messages MUST use HTTP GET.

2.2 Message Syntax

The Query String Response Transfer Protocol extends the **wsignin1.0** message specified in [\[MS-MWBF\]](#) section 2.2. The SAML 1.1 Assertion Extension extends the **SAML assertion** specified in [\[MS-MWBF\]](#) section 2.2.4.2.

2.2.1 XML Namespace References

Prefixes and XML namespaces used in this specification include the following:

Prefix	Namespace URI	Reference
saml	"urn:oasis:names:tc:SAML:1.0:assertion"	[SAMLCore]
adfs	"urn:Microsoft:federation"	This document

2.2.2 Query String Response Transfer Protocol

The Query String Response Transfer Protocol extends the **wsignin1.0** message from [\[MS-MWBF\]](#) section 2.2 to enable passing results in pieces rather than using POST. The protocol does not extend any other message types.

2.2.2.1 wsignin1.0 Message

[\[MS-MWBF\]](#) section 2.2 specifies how parameters are encoded in messages, including new parameters added by extensions. Specifically, it describes that in HTTP GET messages, the parameters are encoded as query string parameters for transmission in the URL. It also specifies how invalid values are handled.

Section [2.2.2.1.1](#) specifies parameters included in both **wsignin1.0** requests and wsignin1.0 responses that use this protocol.

Section [2.2.2.1.2](#) specifies parameters included only in wsignin1.0 responses that use this protocol.

2.2.2.1.1 Common Parameters

ttpindex: The length, in characters, of the **aggregated result** as a 32-bit unsigned integer in decimal notation.

2.2.2.1.2 wsignin1.0 Response

ttpsize: The length in characters of the **pending result** as a 32-bit unsigned integer in decimal notation.

wresult: The current part of the result being transferred, as a string.

2.2.3 SAML 1.1 Assertion Extension

The SAML 1.1 Assertion Extension extends the **SAML assertion** subset as specified in [MS-MWBF] section 2.2.4.2. This extension uses the **SAML advice** element specified in [SAMLCore] section 2.3.2.2 as an extensibility point.

The sections that follow define new elements to convey the source of **claims**, hash data, **user** name, and **SIDs**. These new elements are included as child elements of the advice element in a SAML assertion. These elements use the XML namespace "urn:microsoft:federation". The element content is described using XML schema data types, as specified in [XMLSCHEMA2] section 3.

2.2.3.1 SAML Advice Elements

ClaimSource (optional): A Uniform Resource Identifier (URI) identifying a **requestor IP/STS** or other authentication service (such as a local **account** store) that is the source of the **claims** in the **security token**. The content is of type any URI (as specified in [XMLSCHEMA2] section 3.2.17).

CookieInfoHash (optional): A **base64-encoded** implementation-specific hash value. The content is of type base64Binary (as specified in [XMLSCHEMA2] section 3.2.16).[<6>](#)

WindowsUserIdentifier (optional): A **SID** identifying the **subject** of the **SAML assertion**. The content is of type string (as specified in [XMLSCHEMA2] section 3.2.1) and MUST follow the restrictions for the string representation of a SID, as specified in [MS-DTYP] section 2.4.2).

WindowsUserName (optional): A **user** name associated with the subject of the SAML assertion. The content is of type string (as specified in [XMLSCHEMA2] section 3.2.1) and MUST be of the form "DOMAIN\user name".[<7>](#)

WindowsIdentifiers (optional): A base64-encoded binary structure that defines a set of SIDs that identify the subject of the SAML assertion and a set of flags that specify the use of the SIDs. The content is of type base64Binary (as specified in [XMLSCHEMA2] section 3.2.16), and the binary data MUST be structured as specified in [WindowsIdentifiers Binary Structure \(section 2.2.3.2\)](#).

2.2.3.2 WindowsIdentifiers Structure

The WindowsIdentifiers structure has variable length. It defines a set of **SIDs** and flags. To reduce the overall data size, the SIDs are not included in full binary expansion. Rather, [PACKED SIDS](#) structures are created for each group of SIDs that are identical except for the last subauthority.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
WindowsIdentifierFlags																															
PackedSidsCount																															
PackedSids (variable)																															
...																															

WindowsIdentifierFlags (4 bytes): A 32-bit **WindowsIdentifierFlags** structure (see [2.2.3.2.1](#)).

PackedSidsCount (4 bytes): A 4-byte, **little-endian**, unsigned integer that defines the number of **PackedSids** fields in this structure. This field **MUST NOT** be 0.

PackedSids (variable): A sequence of PACKED_SIDS structures of variable size, each of which defines a set of SIDs. The sequence defines a set of SIDs, which is the union of the sets of SIDs defined by all the elements.

2.2.3.2.1 WindowsIdentifierFlags Structure

The WindowsIdentifierFlags structure is a field of 32 bits.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
WindowsIdentifierFlags																															

WindowsIdentifierFlags (4 bytes): Bits marked 0 **MUST** be 0. The T and N bits operate independently.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	T	N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

T - TryLocalAccount (1 bit): A value of 1 indicates that the SAML NameIdentifier (as specified in [\[SAMLCore\]](#) section 2.4.2.2) takes precedence over the [WindowsIdentifiers structure \(section 2.2.3.2\)](#) element. A value of 0 indicates that the WindowsIdentifiers structure (section 2.2.3.2) element takes precedence over the SAML NameIdentifier (as specified in [\[SAMLCore\]](#) section 2.4.2.2).

N - NoUserSid (1 bit): A value of 1 indicates that a **user SID** is not encoded in the WindowsIdentifiers structure. A value of 0 indicates that a user SID is encoded in the WindowsIdentifiers structure.

2.2.3.2.2 PACKED_SIDs Structure

The PACKED_SIDs structure encapsulates a set of **SIDs** that are identical except for the value of the final subauthority, which is called the **relative identifier (RID)**. The identical portion of the SIDs is included in the **DomainSid** field, and each RID is included separately. The PACKED_SIDs structure has a variable length.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DomainSid (variable)																															
...																															
RidCount																															
RID (variable)																															
...																															

DomainSid (variable): A SID structure of variable size that defines the identical portion of the SIDs encoded in this structure. For details on the SIDs structure, see [\[MS-DTYP\]](#) section 2.4.2.

RidCount (4 bytes): A 4-byte, **little-endian**, unsigned integer that defines the number of RID fields in this structure. This field MUST NOT be zero.

RID (variable): A sequence of 4-byte, little-endian, unsigned integers that define the RIDs for the SIDs encoded in this structure.

2.3 Directory Service Schema Elements

This protocol accesses the following Directory Service schema classes and attributes listed in the following table.

For the syntactic specifications of the following **<Class>** or **<Class><Attribute>** pairs, refer [\[MS-ADTS\]](#), [\[MS-ADA1\]](#), [\[MS-ADA3\]](#).

Class	Attribute
User	All

3 Protocol Details

The following sections specify the **IP/STS** and **relying party** protocol details. Each section details role-specific behavior for the extensions specified in this document. There is not a section for the **web browser requestor** role because additional protocol details for the web browser requestor other than those specified in [\[MS-MWBF\]](#) section 3.4 do not exist.

The IP/STS details (see section [3.1](#)) apply to both the **requestor IP/STS** and **resource IP/STS** roles. The relying party details (see section [3.2](#)) apply to both the resource IP/STS and **Web service (WS) resource** roles.

Because the behavior for issuance and consumption of the **SIDs** is implementation-specific, an abstract data model is not introduced for the SAML 1.1 Assertion Extension. Hence, sections [3.1.1](#) and [3.2.1](#) do not have subsections for that extension.

3.1 IP/STS Details

The following sections specify the protocol details of the **IP/STS**, including details for the Query String Response Transfer Protocol and SAML 1.1 Assertion Extension. These details apply to both the **resource IP/STS** and the **requestor IP/STS** roles.

3.1.1 Abstract Data Model

This section describes a conceptual organization of data that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

3.1.1.1 Query String Response Transfer Protocol

The following sections specify the abstract data model for transmitting the **pending result** as a series of parts in separate messages that are assembled by a **relying party**.

3.1.1.1.1 Pending Result

When the Query String Response Transfer Protocol is used, the result of a **wsignin1.0** action is transmitted by a series of HTTP 302 responses.

The first message in the exchange establishes the **pending result**. Each message exchange in the series transmits a portion of the pending result to the **relying party** until all parts have been delivered. Consequently, the pending result must be available to the **IP/STS** when processing each message exchange until the series is complete.

The message exchange also includes parameters for the current position and the total length of the result (see section [2.2.2.1](#)) to ensure proper construction and error/completion detection. The method used to maintain the availability of the pending result during the series of exchanges is implementation-specific. The IP/STS MUST discard the pending result when cleaning up local state. See [\[MS-MWBF\]](#) section 3.2.5.3.3 for **requestor IP/STS**. See [\[MS-MWBF\]](#) section 3.3.5.4.2 for **resource IP/STS**.[<8>](#)

3.1.1.1.2 Maximum Query String Response Message Length

Although the HTTP protocol (as specified in [\[RFC2616\]](#)) does not place a limit on the length of a URL, some **web browser requestor** implementations have such a limit. **IP/STS** implementations SHOULD use a maximum query string response message length that will allow a broad range of web browser implementations to act as the web browser requestor. This is a limit on the length of the URL after

escaping, including the scheme, authority, path, and query components (as specified in [\[RFC2396\]](#) section 3). The recommended value is 2,083 octets for all messages. For more information, see [\[MAXURL\].<9>](#)

3.1.2 Timers

There are no timers required other than any specified in [\[MS-MWBF\]](#) section 3.1.2; however, a timer MAY [<10>](#) be used to manage state associated with the **pending result**.

3.1.3 Initialization

There are no new initializations beyond any described in [\[MS-MWBF\]](#) section 3.1.3; however, the **pending result** is initialized after the protocol has been initiated (see section [3.1.5.1.2](#)).

3.1.4 Higher-Layer Triggered Events

There are no higher-layer triggered events other than any described in [\[MS-MWBF\]](#) section 3.1.4.

3.1.5 Processing Events and Sequencing Rules

3.1.5.1 Query String Response Transfer Protocol

The following sections specify protocol details for the Query String Response Transfer Protocol when receiving **wsignin1.0** requests.

Sections [3.1.5.1.1](#), [3.1.5.1.2](#), and [3.1.5.1.3](#) specify how **wsignin1.0** requests are processed given the value of the *ttpindex* parameter. Section [3.1.5.1.4](#) specifies common behavior for responding to such requests when the Query String Response Protocol is used.

3.1.5.1.1 Receiving a **wsignin1.0** Request That Does Not Specify a *ttpindex*

When the **IP/STS** receives a **wsignin1.0** request that does not specify the *ttpindex* parameter, the response MUST NOT use the Query String Response Transfer Protocol. The IP/STS MUST discard any **pending result**, and the IP/STS MUST process the message as specified in [\[MS-MWBF\]](#) section 3.1.5.4.

3.1.5.1.2 Receiving a **wsignin1.0** Request That Specifies a *ttpindex* of 0

When the **IP/STS** receives a **wsignin1.0** request that specifies a *ttpindex* parameter value of 0, the IP/STS MUST process the **wsignin1.0** request (as specified in [\[MS-MWBF\]](#) section 3.1.5.4) up to the point where the IP/STS has constructed the **RSTR** element (as specified in [\[MS-MWBF\]](#) section 3.1.5.4.6). The IP/STS MUST apply the following transforms to the RSTR element to produce the **pending result**:

1. Convert the XML string to binary data by applying UTF-8 encoding, as specified in [\[RFC2279\]](#).
2. Compress the result from step 1 to the zlib format (as specified in [\[RFC1950\]](#)) by using the deflate algorithm (as specified in [\[RFC1951\]](#)).
3. Base64-encode the result from step 2, as specified in [\[RFC2045\]](#) section 6.8.

The IP/STS MUST respond as specified in section [3.1.5.1.4](#), by using the result from step 3 as the pending result.

3.1.5.1.3 Receiving a **wsignin1.0** Request That Specifies a *ttpindex* Other Than 0

When the **IP/STS** receives a **wsignin1.0** request that specifies a *ttpindex* parameter value other than 0, the following conditions MUST be evaluated in order:

1. The *ttpindex* value does not conform to message syntax rules (for example, if it is not a number, as specified in section [2.2.2.1.1](#)).
2. The corresponding **pending result** is not available.
3. The value of *ttpindex* is greater than or equal to the length in characters of the pending result.

If the message meets any of the conditions, the IP/STS MUST reject the message and return an HTTP 500 response. Otherwise, the IP/STS MUST respond as specified in section [3.1.5.1.4](#).

3.1.5.1.4 Responding to a wsignin1.0 Request That Specifies a ttpindex

The **IP/STS** MUST construct a **wsignin1.0** response for the **relying party** by using the following values as properly escaped query string parameters in the returned status URL, as specified in [\[WSFederation1.2\]](#) section 13.2.3.

- *ttpindex* MUST be the same value of *ttpindex* as requested.
- *ttpsize* MUST be the same length in characters as the **pending result**.
- *wctx* MUST be the same value of *wctx* as requested.
- *wresult* MUST contain a portion of the pending result, beginning at the character of index *ttpindex* (zero-based), and including as many characters as possible without exceeding the maximum query string response message length. Note that the maximum query string response message length applies after the escaping rules for URL query string parameters.

The IP/STS MUST transmit the message to the **web browser requestor** using HTTP 302 as specified in [\[WSFederation1.2\]](#) section 13.2.3.

3.1.5.2 SAML 1.1 Assertion Extension

The **IP/STS** uses the SAML 1.1 Assertion Extension when responding to **wsignin1.0** requests. Although the protocol details are largely similar, it is necessary in this section to distinguish between the **requestor IP/STS** and **resource IP/STS** roles.

The IP/STS is a requestor IP/STS when issuing a token to a **relying party** that is in a different **security realm** than the IP/STS. Conversely, the IP/STS is a resource IP/STS when issuing a token to a relying party that is in the same security realm as the IP/STS.

3.1.5.2.1 Responding to a wsignin1.0 Request

When responding to a **wsignin1.0** request, the **IP/STS** MAY include any of the [ClaimSource](#) (section [3.1.5.2.1.1](#)), [CookieInfoHash](#) (section [3.1.5.2.1.2](#)), [WindowsUserIdentifier](#) (section [3.1.5.2.1.3](#)), [WindowsUserName](#) (section [3.1.5.2.1.4](#)), and [WindowsIdentifiers](#) (section [3.1.5.2.1.5](#)) elements in the issued **SAML assertion**. For syntax details, see section [2.2.3](#).

The following sections describe the processing semantics for each of these optional SAML assertion elements.

3.1.5.2.1.1 ClaimSource Element

The **IP/STS** MAY [<11>](#) include the ClaimSource element in the issued **SAML assertion**.

3.1.5.2.1.2 CookieInfoHash Element

The **IP/STS** MAY<12> include the CookieInfoHash element in the issued **SAML assertion**.

3.1.5.2.1.3 WindowsUserIdentifier Element

The **IP/STS** MAY<13> include the WindowsUserIdentifier element in the issued **SAML assertion**.

3.1.5.2.1.4 WindowsUserName Element

The **IP/STS** MAY<14> include the WindowsUserName element in the issued **SAML assertion**.

3.1.5.2.1.5 WindowsIdentifiers Element

The **IP/STS** MAY<15> include the WindowsIdentifiers element in the issued **SAML assertion**.

3.1.6 Timer Events

Timer events are not required other than any specified in [MS-MWBF] section 3.1.6; however, a timer event MAY<16> be used to manage the state associated with the **pending result**.

3.1.7 Other Local Events

There are no other local events that impact the operation of this protocol.

3.2 Relying Party Details

These details apply to both the **resource IP/STS** and **WS resource** roles (when acting as a **relying party** for a received **security token**).

3.2.1 Abstract Data Model

This section describes a conceptual model of a possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

3.2.1.1 Query String Response Transfer Protocol

The following sections specify the abstract data model for receiving and concatenating the **aggregated result** as a series of messages.

3.2.1.1.1 Aggregated Result

When the Query String Response Transfer Protocol is used, the result of a **wsignin1.0** action is transmitted by a series of message exchanges. When an **aggregated result** is not available, it is considered to be empty (with size 0). The **relying party** concatenates each portion of the result to the aggregated result when it is received, so the aggregated result must be available to the relying party when processing each message exchange. The method used to maintain the availability of the aggregated result is implementation-specific. The relying party **MUST** discard the aggregated result when it receives a **wsignoutcleanup1.0** message (as specified in [MS-MWBF] section 3.3.5.4.2).<17>

3.2.2 Timers

There are no timers required other than any specified in [MS-MWBF] section 3.3.2; however, a timer MAY<18> be used to manage a state associated with the **aggregated result**.

3.2.3 Initialization

There are no new initializations other than any described in [\[MS-MWBF\]](#) section 3.3.3; however, the **aggregated result** is initialized after the protocol has been initiated (as specified in section [3.2.5.1.3](#)).

3.2.4 Higher-Layer Triggered Events

There are no new higher-layer triggered events other than any described in [\[MS-MWBF\]](#) section 3.3.4.

3.2.5 Processing Events and Sequencing Rules

The Query String Response Transfer Protocol and the SAML 1.1 Assertion Extension introduce new message processing rules.

3.2.5.1 Query String Response Transfer Protocol

The following sections specify protocol details for the Query String Response Transfer Protocol when processing **wsignin1.0** messages.

Section [3.2.5.1.1](#) specifies how to send a **wsignin1.0** request by using this protocol. Sections [3.2.5.1.2](#) and [3.2.5.1.3](#) specify how to process a **wsignin1.0** response given the presence or absence of the *ttpindex* parameter. Section [3.2.5.1.4](#) specifies how the completed **aggregated result** is processed.

3.2.5.1.1 Sending a **wsignin1.0** Request

When using the Query String Response Transfer Protocol, the **relying party** sends a **wsignin1.0** request to the **IP/STS** as specified in [\[MS-MWBF\]](#) section 3.3.5.1 and MUST add the *ttpindex* parameter. When included, the value of the *ttpindex* parameter MUST be the length in characters of the **aggregated result**.

Note If the aggregated result is not available, it is considered to have length 0 (see section [3.2.1.1.1](#)). Therefore, in the first request, the value of *ttpindex* MUST be 0. [<19>](#)

3.2.5.1.2 Receiving a **wsignin1.0** Response That Does Not Specify a *ttpindex*

When the **relying party** receives a **wsignin1.0** response that does not specify the *ttpindex* parameter, the message does not use the Query String Response Transfer Protocol, and the relying party MUST process the message as specified in [\[MS-MWBF\]](#) section 3.3.5.2.

3.2.5.1.3 Receiving a **wsignin1.0** Response That Specifies a *ttpindex*

When the **relying party** receives a **wsignin1.0** response that specifies the *ttpindex* parameter, it MUST evaluate the following conditions in order:

1. The *ttpindex* value does not conform to message syntax rules (for example, it is not a number; as specified in section [2.2.2.1.1](#)).
2. The *ttpindex* parameter is not equal to the length, in characters, of the **aggregated result**. Note that if the aggregated result is not available, it is considered to have length 0 (as specified in section [3.2.1.1.1](#)).

If the message meets one of these conditions, the relying party MUST reject the message and return an HTTP 500 response.

The relying party MUST construct a new aggregated result. If *ttpindex* is 0, the aggregated result MUST be the value of the *wresult* parameter; otherwise, the aggregated result MUST be constructed by appending the *wresult* parameter to the current aggregated result.

The relying party MUST evaluate the *ttpsize* parameter as follows:

- If the length in characters of the new aggregated result is greater than the value of the *ttpsize* parameter, the relying party MUST reject the message and return an HTTP 500 response.
- If the length, in characters, of the new aggregated result is equal to the value of the *ttpsize* parameter, the relying party MUST process the completed result as specified in section [3.2.5.1.4](#).
- If the length, in characters, of the new aggregated result is less than the value of the *ttpsize* parameter, the relying party MUST request the next portion of the result as specified in section [3.2.5.1.1](#).

3.2.5.1.4 Processing the Complete Aggregated Result

When the **aggregated result** is complete, the **relying party** MUST apply the following transforms to the aggregated result to produce an XML string:

1. Base64-decode the aggregated result to binary data (as specified in [\[RFC2045\]](#) section 6.8).
2. Decompress the result from step 1 from the zlib format (as specified in [\[RFC1950\]](#)) by using the inflate algorithm (as specified in [\[RFC1951\]](#)).
3. Interpret the binary data from step 2 as a UTF-8-encoded string (as specified in [\[RFC2279\]](#)).

If any error occurs when applying the transforms, the relying party MUST reject the message and return an HTTP 500 response.

The relying party MUST process the **wsignin1.0** response (as specified in [\[MS-MWBF\]](#) section 3.3.5.2), substituting the string output from step 3 for the *wresult* parameter.

3.2.5.2 SAML 1.1 Assertion Extension

The method of evaluating **SIDs** transmitted by using the SAML 1.1 Assertion Extension is implementation-specific. For specifications about SIDs and their semantics in Windows, see [\[MS-DTYP\].<20>](#)

3.2.6 Timer Events

There are no timer events required other than any events specified in [\[MS-MWBF\]](#) section 3.3.6; however, a timer event MAY [<21>](#) be used to manage a state associated with the **aggregated result**.

3.2.7 Other Local Events

There are no other local events that impact the operation of this protocol.

3.3 Web Browser Requestor Details

This section specifies the **web browser requestor** role in transporting protocol messages.

3.3.1 Abstract Data Model

A **web browser requestor** does not need to understand any protocol-specific data for the correct operation of the protocol. It MUST be able to support HTTP query string and POST body parameterization. To provide the best end-user experience, it SHOULD be able to support HTTP cookies (for more information, see [\[RFC2965\]](#)).<22>

3.3.2 Timers

A **web browser requestor** does not depend on timers beyond those that are used by the underlying transport to transmit and receive messages over HTTP and SSL/TLS, as specified in section [3.1.2](#).

A web browser requestor does not need to be aware of an implementation's use of timers to determine when the validity intervals of **security tokens** and authentication contexts expire.

3.3.3 Initialization

There is no protocol-specific initialization for a **web browser requestor**. It only needs to be ready to perform the standard HTTP 1.1 methods required for accessing **WS resources**. Specifically, it **MUST** support HTTP GET and POST methods and properly respond to HTTP 1.1 redirection and error responses.

3.3.4 Higher Layer Triggered Events

Protocol messages are exchanged between a **requestor IP/STS** and a **relying party**. The only function of the **web browser requestor** with respect to the protocol is to transport these messages. The web browser requestor can be triggered to begin protocol message exchange by receipt of an HTTP/1.1 302 found that includes a location directive, or an HTTP/1.1 200 OK that includes a form with method set to POST.

3.3.5 Processing Events and Sequencing Rules

A **web browser requestor** plays a passive role in the operation of the protocol. Its only function is to transport protocol message requests and responses between a **requestor IP/STS** and one or more relying parties. It is not required to understand the types or content of these protocol messages. The web browser requestor **SHOULD** transport all protocol message requests and responses between a requestor IP/STS and a **relying party** without changing the messages at all. [<23>](#)

3.3.6 Timer Events

A **web browser requestor** does not need to interact with any timers, or service any timer events, beyond those that might be used by the underlying transport to transmit and receive messages over HTTP and SSL/TLS, or those specified in section [3.1.6](#).

3.3.7 Other Local Events

A **web browser requestor** does not have dependencies on local events beyond those specified in section [3.1.7](#).

4 Protocol Examples

4.1 Query String Response Transfer Protocol

4.1.1 Annotated Example

The following is a protocol example for the Query String Response Transfer Protocol.

The Query String Response Transfer Protocol is best understood as occurring abstractly between an **IP/STS** and a **relying party**, because the changes to the Web Browser Federated Sign-On Protocol [MS-MWBF] are applied consistently whether between **requestor IP/STS** and **resource IP/STS** or between resource IP/STS and **WS resource**.

This annotated example shows a Query String Response Transfer Protocol exchange between a requestor IP/STS and a resource IP/STS. It is part of a larger network trace (see section 4.2) that also uses the Query String Response Transfer Protocol between the resource IP/STS and the WS resource.

The following table specifies the protocol roles of the hosts.

Protocol role	Host name
IP/STS	adatumsts-7
Relying party	treysts-7

Each HTTP message is prefaced by an annotation that describes its recipient and purpose. This annotated example omits many elements of the HTTP messages. For example, implementation-specific cookies and superfluous HTTP headers are not included. The full messages are specified in section 4.1.2. The following parameters are specified in this document and appear in the HTTP messages that follow:

- *ttpindex*
- *ttpsize*
- *wresult*

The following are the processing steps of this annotated example:

1. Just prior to the example, the **web browser requestor** made a GET request to the relying party (treysts-7).
2. The relying party returns an HTTP 302 message that specifies a **wsignin1.0** request for the IP/STS. This wsignin1.0 request includes the *ttpindex=0* parameter, which initiates the Query String Response Transfer Protocol (see section 3.2.5.1). At this time, the relying party's **aggregated result** is empty, as specified in section 3.2.5.1.1.

```
HTTP/1.1 302 Found
Location:
https://adatumsts-7/adfs/ls/?wa=wsignin1.0&
wtrealm=urn%3afederation%3atreys+research&
wct=2006-07-13T07%3a32%3a21Z&
wctx=https%3a%2f%2ftreys-test%2fclaims%2f%5chttps%3a%2f%
2ftreys-test%2fclaims%2fDefault.aspx&ttpindex=0
```

3. The web browser requestor relays the wsignin1.0 request to the IP/STS (adatumsts-7) in an HTTP GET message.

```
GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation
%3atreys+research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%
3a%2f%2ftreys-test%2fclaims%2f%5chttps%3a%2f%2ftreys-test%
2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1
```

4. The IP/STS engages in a series of messages outside the scope of the protocol whereby it ascertains the **user** identity. These messages are omitted, as they have no bearing on the Query String Response Transfer Protocol. For more details, see [MS-MWBF] section 3.1.5.4.3.
5. Once the user's identity has been determined, the IP/STS creates an **RSTR**, as specified in [MS-MWBF] section 3.1.5.4.6. The RSTR is transformed into the **pending result**, and the first portion is returned in a wsignin1.0 response (as specified in section 3.1.5.1.4). In this message, the **wresult** parameter is the first 1,727 characters of the pending result. The **ttpsize** parameter indicates the length of the full pending result, 2,652.

```
HTTP/1.1 302 Found
Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&
ttpsize=2652&ttpindex=0&wctx=https%3a%2f%
2ftreys-test%2fclaims%2f%5chttps%3a%2f%2ftreys-test%2fclaims%
2fDefault.aspx&wresult=eNrNWftzqsos%
2fUW6zGVBYg3rCRlhquAaAlyssphOEShVEGRpz1e9BoTNYt59R62D4NPW3311%
2f3dDM8lDgFWHBXQJzb0CuyOK%2bmaA1TC%2bItSjFshJJNigde7bEZ5f12QFHYi2Di
4u9kByN3%2bx1lIVViqkXTHYpuUXlW4Lz59HBjGfofbD89YDfZDADGMMtjlDauK1V8
bP436NAMGzD%2bvddps%2fftFuzd91ewc79iu5y7Yv1%2b4HeadRXjAqopzt2UQCPOu
%2fd0755hp3RvwLYGrZ7zppM9NossHQTQh51bOxm4vpsXSbNhuK8om8MME
%2bFjkyGCOP0oOAdfoz0bQS6O8SB1E4gHuTEwTEaMN
%2fpGxsJoPkWm4BSP64FuDFGOQ8D1MFfoSQKk3SSgSCvsd7q9C86F8YKP4apB0ly8i
z2avtXR59Unj7FnGewamQQzFzozgfo%2b6n55%2bbpz4FdnHo7
%2bPahOsHmOyidUyyEqChi6M3%2buqdM31J7GUIoyC%2fQdZ8UqK5vj2Kulay0tQy
KoWL1HhnM8%2fS%2bvGB%2bondK94312%2fg8wimeeydrNq5m8OEPDc%2byv9QMB%
2bvDZhhYp%2bxfD4oDco49VGJLzmyi9Ur9PK3pzEpFdWvtYMYZg0ZZYn7
%2b5Pkbdw4wdTsedx8An4SpzFJhpuj7D%2fnsV3uoeQt%2fo%2fGL6xcAVC
%2f4%2bXCWk4yvSpy%2bHnj3xPGR5yN66r%2b22NTStx4A3yflDZuftzEW9eDXwDZ
%2f0xj7m4K%2bOT%2bHvUn7c%2fSP%2bEWUJKgtF7%2fXdQGUwydKL87yBVmLRs
%2fy5Iod4kCU7lInID9d8ME222G919HSf3yXNlXmJiiyt5m6hVXWZbfs
%2fYeh3QlqmI5iij4OA6%2fNc%2f%2fgn7d%2bp4eBDdFKTnJm
%2fh406AaYBMiMlyj5BcmGYgha5P38ODde0w7%2fdZsUDd4vmjma7IMu
%2fc4cpmTJQsGMKsnR2NmQY%2fNb18b4U8P08xNMRmNCb5Z%2f284YLqHG7SF
%2fj2%2bhHOC9HVZv2CHugUnxiGzif8PTleOzibOVTPhN2GVHqWsG5S6BHgV4f3E
0eycpnEu9V8k07ckvVtLVyzdlYUYTg1clGn%2fJ5011Fmx1mmaTOBae%2bMEojVHb
%2bd96NDIj09jLvrlopNJXwsPNN3kWWtuE2Vj4t90Zram4XtYg1Y2jTyLGrUFmYHmd
V4WW87i6VIm9VO3UNaBPOiR5XrV0emvSA5kpm9SXZK7rXmHZ
%2fGWXdBd5mpuJX07ZiQ4ru01X%2fmi6gf4De4zmQG
%2fAPOqz0US06Nce6uXteCFVLVVCPLtLYVfVkreEAWRWCuuVBqOoKGyayCeqfdJiCMR
%2bud9E6Vris5oE5k4EIHMPyStlcinPTFKWYm3OVTeXiFOSlVmeWcHtfBFAu6XL8C
lhjqlbGFJSGOHZPsnNFJplkBr88yCKw%2bXA854ExFVryermwmNViTHsVzRqQehiA
VgR7p9jqihVnQcYBQFSZA1HgY1PnQ10017ue4uB4smhLvZZth3SxY7QX1thxeQF0p
%2b9W%2bNmJNkfjZb%2fTga7DDMWjyOeTmeZ07vb8EogLjTrK%2bZRq6weJ1WNn02
k9F6RVHa3%2bWqcZXRkto8gHGZR5oZfYq4g72IsKhFxVavFycmf0Eas4dKvg9xCec
SI0h1rSDYWDpzG7tcUxxz7Q5WOpisAEPGqrPD0S7p1%2bKYAES4t%2bBuaQIsGJIJ
yWhmTexqyJpcka4qzWVXX
```

6. The web browser requestor relays the wsignin1.0 response to the relying party (treysts-7) in an HTTP GET message.

```
GET /adfs/ls/?wa=wsignin1.0&
ttpsize=2652&ttpindex=0&wctx=https%3a%2f%2ft
reys-test%2fclaims%2f%5chttps%3a%2f%2ftreys-test%2fclaims
%2fDefault.aspx&wresult=eNrNWftzqsos%2fUW6zGVBYg3rCRlh
quAaAlyssphOEShVEGRpz1e9BoTNYt59R62D4NPW3311%2f3dDM8lDgFWHBXQJzb0
CuyOK%2bmaA1TC%2bItSjFshJJNigde7bEZ5f12QFHYi2Di4u9kByN3%2bx1lIVViq
kXTHYpuUXlW4Lz59HBjGfofbD89YDfZDADGMMtjlDauK1V8bP436NAMGzD%2bvddps
%2fftFuzd91ewc79iu5y7Yv1%2b4HeadRXjAqopzt2UQCPOu%2fd0755hp3RvwLYGr
```

Z7zppM9NossHQTh51bOxm4vpsXSbNhuK8om8MME%2bFjkyGCOP0oOAdfoz0bQS608
 SB1E4gHuTewgTEaMN%2fpgXsJoPkWm4BSP64FuDFGOQ8D1MFfoSQKk3SSgSCvsd7q9
 C86F8YKP4apB01y8iz2avtXR59Unj7FnGewamQQQzfzofgqo%2b6n55%2bbpz4FdnH
 o7%2bPahOsHmOyidUyyEqChi6M3%2buqdm31J7GUIOyC%2fQdZ8UqK5vj2Kulay0tQ
 yKoWL1HhnM8%2fS%2bvGB%2bondK94312%2fg8wimeeydrNq5m8OEPDc%2byv9QMB%
 2bVDZhHyP%2bxfD4oDco49VGJLzmyi9Ur9PK3pzEpFdWvtYMYZg0ZZYn7%2b5Pkbaw
 4wdTsedx8An4SpzFJhpuj7D%2fnsV3uoEQ%2fo%2fGL6xcAVC%2f4
 %2bXCWk4yvSpy%2bHnj3xPGR5yN66r%2b22NTStx4A3yflDZuftzEW9eDXwDZ
 %2fOxj7m4K%2bOT%2bHvUn7c%2fSP%2bEWUJKgtF7%2fXdQGQUwydKL87yBVM1Rs
 %2fy5Iod4kCU7inID9d8ME22G919HSf3yXN1xmJiIyt5m6hVXWZbfs
 %2fYeh3QlmgI5iij4OA6%2fNc%2f%2fgn7d%2bp4eBdDFKtNjM%2fh406AaYBMiM1
 yj5BcmGXqha5P38ODde0w7%2fdZsUDd4vmjma7IMu%2fc4cpmTJQsGMKsnR2NmQY
 %2fNb18b4U8P08xNMRmNCb5Z%2f284YLqHG7SF%2fj2%2bhHOC9HVzv2CHugUnxi
 GZiF8P1eOzibOVTPHN2GVHqWsg5S6BHgV4f3E0eycpnEu9V8oK7ckvVtLVyzd1
 YUYTg1c1Gn%2fJ5011Fmx1mmaTOBae%2bMEojVHb%2bd96NDIj09jLvrlopNJXws
 PNN3kWwtUE2Vj4t90Zram4XtYglY2jTyLGrUFmYHmdV4WW87i6VIm9VO3UNaBPOi
 R5XrV0emvSA5kpm9SXZK7rXmHZ%2fGWXdBd5mpuJX07ZIQA4ru01X%2fmI6gf4De
 4zmQG%2fAPQz0US06NCE6uXteCFVLVVCPLtLYVfVkrEEAWRWCuVBqOoKGyayCe
 qfdJiCMR%2bud9E6Vris5oE5k4EIHMPyStlcinPTFKWYm3OVTeXiFOS1VmeWcHtf
 BFAu6XL8ClhjqlbGfJSGOHZPsnFJplkBr88yCKw%2bXA854ExFVryermwNViTH
 sVzRqQehiAVGr7p9jqihVNqcYBQFsZA1HgY1PnQ10017uE4uB4smhLvZZth3SxY7
 QX1thxeQF0p%2b9W%2bNmJNkfjZb%2fTGa7DDMWjyOeTmeZO7vb8EogLjTrK%2bZ
 Rq6weJlWNn02k9F6RVHa3%2bWqcZXRkto8gHGZR5oZfyQ4g72IsKhFxVavFycmf0E
 as4dKvg9xCeC5IOh1rSDYWDpzG7tcUxxz7Q5WOpisAEPGqrPDos7p1%2bKYaES4t%
 2bBuaQISGJIJyWhmTexqyJpcka4qzWVXX HTTP/1.1

7. Because the relying party's aggregated result is empty, the *wresult* parameter becomes the aggregated result. Because the length of the *wresult* parameter was 1,727 characters, which is less than the length of the *ttpsize* parameter of 2,652 characters, the relying party needs to request more data from the IP/STS. It does this by sending an HTTP 302 with a *wsignin1.0* request that includes the *ttpindex=1727* parameter. For further specifications, see section [3.2.5.1.3](#).

HTTP/1.1 302 Found
 Location: https://adatumsts-7/adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atrety+research&wct=2006-07-13T07%3a32%3a27Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=1727

8. The web browser requestor relays the *wsignin1.0* request to the IP/STS (adatumsts-7) in an HTTP GET message.

GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atrety+research&wct=2006-07-13T07%3a32%3a27Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=1727 HTTP/1.1

9. The IP/STS returns the remaining 925 characters of the pending result, beginning with character 1,727. For further specifications, see section [3.1.5.1.3](#).

HTTP/1.1 302 Found
 Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&ttpsize=2652&ttpindex=1727&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&wresult=kqNHeG5OcCSAUgEHIEISLriGRPIQJBEM%2bnGViaAj8ERCSwkseRfLs13wPzbYkh%2bZs6Vn0YZP76zacMJFGrZbrVe%2fYXvUjtZTKE7YND8pSqH2XS543Z0Piuxye9155PxlBGZyO6WHThaJc9Xt2NPJctVWbL2tsZNZOTdPugEvtcupyOtIikMLOZI7tGhPRPtRa7wh9RP5CwuNEqvwqs7ritXKUcJsl6y291pc4gtElqL3Zmte%2bdK88BIOrwTux%2fojel5SlqGDCC9YgeuWLyWeV5UxqsfdkNR3%2breIOauD%2fKFVtUL%2bflnbVmTGV8LPf16dhUYvF3PalX%2fhMzVLC7xzZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2Xykh791vFMvVaOs1lrkk

```
%2fuCc7LnmQTigP58D%2fnQOyLk1eXYX3x3GU8rrUsn84LO%2bcGRAtt4eNHY1VD
NEDcGh1%2fYOCujud%2fup3%2bZefDPvB3s91IYB7%2b3VXuK%2bwLyUR23Le07Y
WTLPlDl1p62FeOiHAQuF53jLdrGtc%2b3jPFPkwJFdrk%2b9HlC7ZNDdiyLPcivf
SV2bCmN6ZEmxMRJ0z3KS6G426mldqL0MYcV1WJ3zzJHR8pWVozlral0wbkuYqT0l
eD60fT1hBCVDicNLlmwCRiKRIAu9QjsGoiff4T63Xg3zIFiq6rSQ%2fRvr29y6LF
oyBn2NHQ6T1oTaG%2f1lJe46h3nqJM8dbElFMpNTRKW7SYduLSjlaHXZKO2m0
%2fTykrVn3ny93nGygPpVKClJt3qQfFAfe6YZ8m5m1LXDvvee6%2fvKpfLKxH8
7tJe4u2AtJhNDPEUXb8LbL%2fwXaBdT8Qt2sRedfow4A6k1N%2biOM3f3xwu9ty
v2euTi%2bHppT9Owzebb7eAP9x8bzXPTz9gOclvYv3Iy08%2fkzz9A6cx%2fkW
%3d
```

10. The web browser requestor relays the `wsignin1.0` response to the relying party (treysts-7) in an HTTP GET message.

```
GET /ads/ls/?wa=wsignin1.0&
ttptime=2652&ttptimeindex=1727
&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a
%2f%2ftreyws-test%2fclaims%2fDefault.aspx&
wresult=kqNHeG5OcCSAUgEHIEISLriGRPIJQBEM
%2bnGViaAj8ERCSwkseRfLs13wPzbYkh%2bZs6Vn0YZP76zacMJFGrZbrVe
%2fYXUvJtZTKE7YND8pSqH2XS543Z0Piuxye9155PixlBGZy06WHThaJc9Xt2NP
JctVWbL2TsZNZOTdPugEvtcupyOtIikMLOZI7tGhPRPtRa7wh9RP5CwuNEqvwqs
7ritXKUcJsl6y291pc4gtElqL3Zmte%2bdK88BIOrwTux%2fojel5SlqGDCC9Y
geuWLyWeV5UxqSfdkNR3%2bREI0auD%2fKFVTuL%2bflnbVmTGV8LPfl6dhUYvF
3PalX%2fHmzVLC7xzZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2Xykh791vFMVaO
sllrkk%2fuCc7LnmQTigP58D%2fnQOyLk1eXYX3x3GU8rrUsn84LO%2bcGRAtt4
eNHY1VDNEDcGh1%2fYOCujud%2fup3%2bZefDPvB3s91IYB7%2b3VXuK%2bwLyU
R23Le07YWTLPlDl1p62FeOiHAQuF53jLdrGtc%2b3jPFPkwJFdrk%2b9HlC7ZND
diyLPcivfSV2bCmN6ZEmxMRJ0z3KS6G426mldqL0MYcV1WJ3zzJHR8pWVozlral
0wbkuYqT0leD60fT1hBCVDicNLlmwCRiKRIAu9QjsGoiff4T63Xg3zIFiq6rSQ
%2fRvr29y6LFoyBn2NHQ6T1oTaG%2f1lJe46h3nqJM8dbElFMpNTRKW7SYduLS
jlaHXZKO2m0%2fTykrVn3ny93nGygPpVKClJt3qQfFAfe6YZ8m5m1LXDvvee6
%2fvKpfLKxH87tJe4u2AtJhNDPEUXb8LbL%2fwXaBdT8Qt2sRedfow4A6k1N
%2biOM3f3xwu9tyv2euTi%2bHppT9Owzebb7eAP9x8bzXPTz9gOclvYv3Iy08
%2fkzz9A6cx%2fkW%3d HTTP/1.1
```

11. The relying party appends the `wresult` parameter to the aggregated result. The new aggregated result is 2,652 characters long, which indicates the completion of the Query String Response Transfer Protocol (see section 4.1). The relying party extracts the RSTR from the aggregated result (as specified in section 3.2.5.1.4) and processes it as specified in [MS-MWBF] section 3.3.5.2. The relying party's next action is outside the scope of the protocol, though in this case the relying party, which was a resource IP/STS, issued a new **SAML assertion** and used the Query String Response Transfer Protocol to transmit it to a WS resource.

4.1.2 Full Network Trace

The following is the full network trace for the protocol example in [Annotated Example \(section 4.1.1\)](#).

The following table specifies the protocol roles of the hosts.

Protocol role	Host name
Requestor IP/STS	adatumsts-7
Resource IP/STS	treysts-7
WS resource	treyws-test

HTTP requests are prefaced by a line with three right-angle brackets ("**>>>**") and the name of the host to which the request was sent. Requests are followed by the HTTP response, which is prefaced by

a line with three left-angle brackets ("<<<"). The final three HTTP messages (HTTP 302, HTTP GET, and HTTP 200 OK) are an internal implementation detail and shown for completeness only. These messages are not necessary for interoperability.

```
>>> treyws-test (WS Resource)
GET /claims/ HTTP/1.1

<<<
HTTP/1.1 302 Found
Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&wreply=https%3a%2f%2ftreyws-test%2fclaims%2f&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0

>>> treysts-7 (Resource IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wreply=https%3a%2f%2ftreyws-test%2fclaims%2f&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1

<<<
HTTP/1.1 302 Found
Location: https://adatumsts-7/adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0
Set-Cookie: _TTPRealm=urn:federation:adatum; path=/adfs/ls; secure; HttpOnly

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1

<<<
HTTP/1.1 302 Found
Location: /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1

<<<
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Negotiate
WWW-Authenticate: NTLM

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1
Authorization: Negotiate TlRMTVNTUAABAAAAB4IIogAAAAAAAAAAAAAAAAAAAA
FAs4OAAAADw==

<<<
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Negotiate TlRMTVNTUAACAAAEAAQAdgAAAAFgomi1Z0tC7za4
J0AAAAAAAAAAAAQBBAFIAAAABQL0DgAAAA9BAEQARgBTAFYATQataEEAAgAQAEARABGAF
MAVgBNAC0AQQABABYAQQBEAEAVABVAE0AUwBUAFMALQA3AAQAQgBhAGQAZgBzAHYAbQA
tAGEALgBuAHQAdABlAHMAdAAuAG0AaQBjAHIAbwBzAG8AZgB0AC4AYwBvAG0AAwBSAGEA
ZABhAHQAdQBtAHMAdABzAC0ANwAuAGEAZABmAHMAdgBtAC0AYQAuAG4AdAB0AGUAcwB0A
C4AbQBpAGMAcgBvAHMAbwBmAHQALgBjAG8AbQAFADoAYQBkAGYAcwB2AG0ALQBhAC4Abg
B0AHQAZQBzAHQALgBtAGkAYwByAG8AcwBvAGYAdAAuAGMAbwBtAAAAA=
```

```
>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys+research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreys-test%2fclaims%2f5chttps%3a%2f%2ftreys-test%2fclaims%2fDefault.aspx&ttindex=0 HTTP/1.1
Authorization: Negotiate TlRMTVNTUAADAAAAGAAAYIAAAAAAYABgAmAAAAABAAEABI
AAAAAGgAaAFgAAAAOAA4AcgAAAAAAACwAAAAABYKIoGUCzg4AAAAAPYQBkAGYAcwB2AG0AL
QBhAGEAZABtAGkAbgBpAHMAdABYAGEAdABvAHIASgBTAEIALQBEAEUAVgBRfVjBrWkBSQ
AAAAAAAAAAAAAAAAAADZmf/wdoShwGwc7CBCpFNGdUHCLsDZPUU=
```

```
<<<
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Negotiate
WWW-Authenticate: NTLM
```

```
>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys+research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreys-test%2fclaims%2f5chttps%3a%2f%2ftreys-test%2fclaims%2fDefault.aspx&ttindex=0 HTTP/1.1
Authorization: Negotiate YIIFRwYgKwYBBQUCoIIFOzCCBTegJDAiBgkqhkiC9xI
BAGIGCSqSgIb3EgECAGYKKwYBBAGCNwICCqKCBQ0EggUJYIIIFBQYJKoZIhvcSAQICAQB
uggt0MIIIE8KADAgEFOQMAQ6iBwMFACAAACjggQDYYID/zCCA/ugAwIBBAEfGx1BREZ
TVk0tQS5OVFRFUF1QuTuU1DUk9TT0ZULkNPTaIeMBygAwIBAqEVMBMBBehUVFABc2FkYXR
lbXN0cy03o4IDstCCA62gAwIBF6EDAgECooIDnwSCA5syDvJHfsRntPF0oUy00/THHoA
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Dj3e3uQffYfLuw5AFkxDJlby+q/TBaOtwq/kUfWKTcg1TziWl3wNyj3CmShctPXlxsJU
SUMJj5uVYm9gl7cv0CnaCRM4TJVlQd+eWcuXlwcMrujg2ktZZHAgmk8VxfqjrUyWJbw
OGqalZ60uxTId9k3Rxt1kwBICtWgPgeHsDQ3nMOPpHXFlTNqPQvYmKOnCNeJAsqAJ34m
OHW4D/Q3qfj9P1fDw1fxolDBM6A1UJ/rpXHGyV3z5EJKLPYqRYPyAm9hrWmmu4/SIA1r
+5rZHWb5vTuh4jLiHDGb12ReQIPhMvP6CUVJGggbIjyvrX5VoYCIIm+uRPP3GFej6C4eX9
fMrwqN3ktk76bRWZf9HKb9VHuYX/k+tMlktf2VrYAifyQf8oJxMq9XON+Iej3KvqyP3d
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mVzS2WrZVFWzk2fufFovcgxHc5hzuWav4VtxD0KWp1UDnr8DmAK3T65dJ8AsgKpfKaxn
vbZ0+0jDWFZVErgGqgnJCefuOU9QDm/oAftGgaruQrAUdeAJVUrfNE6rott6jMXECVIO
IxtDbAtugbtNhw/9f61Tc0xY+7y/tmY+N5MK1Y/my8a/jJ0+DjFR1UJXUWJMDKun0ITt
lxaAmjJAmGsgkmkhV6MoAd7JDq3je3oogKcz2rfAavRZt/Gy8Zhvb+QKNFeMWpKq7wXm
fUa9w/z6V6q3VHomzipcSGUkr2RQrRpnTcIBopGZcy/kbV1G9RqSB0zCB0KADAgEXooH
IBIHFGfGfTuh/3yxOaGfythDxT33t4KCBiNzQ/SLH5M/iH4oxfgtJNbJ5tNM3rXlRkY+
36tO0GgLfJUTaaBS5P0Fms5lHir9PbK4nY7n335HcDVJQXBfiudOkqGU3U80Kuh6UPrS+
ihacMygHCR5E0lDgl1tqvEObraoRR0ct1EvnmMBQG6sajBRt800sqEjuldl77U7W0Aa9I
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```

```
<<<
HTTP/1.1 302 Found
Date: Thu, 13 Jul 2006 07:32:27 GMT
Server: Microsoft-IIS/6.0
X-Powered-By: ASP.NET
WWW-Authenticate: Negotiate oYGgMIGdoAMKAQChCwYJKoZIgvcSAQICooGIBIGFYI
GCBgkqhkiG9xIBAGICAG9zMHGgAwIBBAEDAgEPomUwY6ADAgEXolwEWjj9oljcrPxx9ipQ
kjQo69bf5SYXFD7mxzAlpl8q5jKcV4ETZcXVawsYxvnHGV/wTsl/yg8CHMZ8D07cNqsp
IEUkG6joNvB9NqGa4q5441JbEVAblZPZ/9w==
Location: https://treys-7/adfs/ls/?wa=wsignin1.0&ttsize=2652&ttindex=0&wctx=https%3a%2f%2ftreys-test%2fclaims%2f5chttps%3a%2f%2ftreys-test%2fclaims%2fDefault.aspx&wresult=eNrNWFTzqsoS%2fIUW6zGVBYg3rCRlhq
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```

```
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research; path=/adfs/ls/; secure; HttpOnly
Set-Cookie: TTPDest=urn:federation:trey research; path=/adfs/ls/;
secure; HttpOnly
Set-Cookie: TTPData=eNrNWftzqsos/iuW6zGVBYg3rCrlhquAaAAlyssphOEShVEGR
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```

```
>>> treysts-7 (Resource IP/STS)
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pisAEPGqrPDoS7p1%2bKyAEs4t%2bBuaQtsGJIjYWhmTexxyJpcka4qzWVXX HTTP/1.1
Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg,
application/vnd.ms-excel, application/vnd.ms-powerpoint,
```



```
application/msword, application/x-shockwave-flash, */*
Accept-Language: en-us
UA-CPU: x86
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.2;
SV1; .NET CLR 1.1.4322; .NET CLR 2.0.50727; InfoPath.1)
Host: treysts-7
Connection: Keep-Alive
Cookie: _TTPRealm=urn:federation:adatum

<<<
HTTP/1.1 302 Found
Date: Thu, 13 Jul 2006 07:32:27 GMT
Server: Microsoft-IIS/6.0
X-Powered-By: ASP.NET
X-AspNet-Version: 2.0.50727
Location: https://adatumsts-7/adfs/ls/?wa=wsignin1.0&wtrealm=urn%
3afederation%3atrey+research&wct=2006-07-13T07%3a32%3a27Z&wctx=
https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test
%2fclaims%2fDefault.aspx&ttindex=1727
Set-Cookie: _TTPData=eNrNWftzqsoS/iuW6zGVBYg3rCRlhquAaAA1yssphOEShVEG
RPz1e9BoTNYt59R62D4NPW3311/3dDM81DgfWHBXQJzb0CuyOK+maA1TC+ItSjFsHJJNi
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IJyWhmTexqyJpcka4qzWVXX; path=/adfs/ls/; secure; HttpOnly

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atrey+research&
wct=2006-07-13T07%3a32%3a27Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%
2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttindex=1727
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>>> treys-test (WS Resource)
GET /claims/Default.aspx HTTP/1.1
Cookie: _WebSsoAuth=eNrNV9uSosoS/RXDeTR6wLsY3R2nAEFQUBAVfdlRQnFRqViuIn79LrL1unvOzOmzYx62T0WYKnPlyiSrfE5gtO+DJEfXGhJceawU8ax6V7fTQM32pv1U73bZp5bndJ64Taf+xPVYB0EPOe6mVa0oSziHBScpXOlLtcGynSe2+1RvWmy332z0G731GyZ+qWYx7nvIRTEsg/TTGBWVGUCIxxk5QrWhws+IFihP67qVap4YQfzScoz10+iXpmy8CkzDpYxiHJ86/RnQxv36d7YP731UX5+vKQoEu2FpSCo6SXnkkRj9iifFTPakB15aUn6P6d0xb15B5oYIO8hESRqHTun/EegT5DVI00PSZ5gy5zx5SukWxtndMEqYZ+Yj9NPzf/fOfMrrHs89haUL6HpJXyj9z0gWO+hNutJ8ky4KnZgkxEvfFaT6+q1A0IVpFj0zn73d/ROyC2ntPTKESfDfEA3DiTkoSt5JNzIrnRfnYCCIC5f02Jd7pA9+H3q8pfYmThognIb0lessHSmK6HPlo/03bdbl1f4I1lAaEPenJoUfQP08xC7Jk3sLzLLNFjnp25NOO1FxS7QXorgikTiCNHpZe1r6xAlQBJPvVKEEWMN3EvV3FphP9eorcKMqH7TYMCXxf27af3dI9Jb/R+d3VR4EmN/pclctPz20yVL0+cW/J42PPCuPVBntPqIYLgHrkuHRL9+DI5QAd9gWT1c4wF3GfoFF6e9Sf0Z+v/4i2QKC4XP9Z1hplTCT01fzPMJVjkh3+LMmyfyrgcIjJCcX/bpa0DaMwTb9Ok/nldzULfUybKH4bVq9ieZ5/z5tXpNqqsQzLMRTgJqH/rXrbhdxy9L0+CxATTl/kfXh5N6AqYO+TOEyD6Bcu60ydlV0+obPz5NRb+Fu1wrzj80U3H5jFCXxKali/eyJkRh+LyZKrmTEw1+ulrF4XXZyuGOKEnb5S8W/9/PBA+oT05IPcpuadzpfRld79Qh31PTgx9eub+E50eGt1c3LrGxS37XJsQ1FOHY3CcYXGs13G6z6YOPfHeI5+Zh7Z0/b4XhtW7AbEoL4iJrQsO19G8FWPjEOXucHh2zoXm6b12Fy4necauw8UiWlmtg7NkkLbbLpVeMrLV4Q4m7cxVetM4thbpOVoyulYuadla4HusGhSzaD2FKfZCLMUuS6wPU7X22jbHk1mzjrzXXTW4GEqNmlpxm2M5BwYG7VptxL2NLnwh5odKbY6VSDb7eWWyDvyzyNU3LKy2ywnwhTEvKJ5Z/PK04sONkUR2a0ggPjog1zhga+M5JakNn1Q/gbnC9B5f3cMdgHM5SwPjLkERLDWTCeXjJW4MAxxkLfnUN4Xa4kPHGy25xF3ckWApJzn9S1oapZSaNYg10QVxm2Xu0252jTeOA+2wOB9fcEDzRjss1jb+4s71FL9IvD4Jw4DCYAjvzVA7q92gr8agCwWongugNAfBkatmOb8dCeg1SXNQJTJhpJpS4HRyYepDcLRqegbl3qlFNNyLC21sKWtuNEensSYtJj14KfY1enUNJyPNWY4XE9UeTlIG50tna7enEnbcoYGAxq0xDLRY2XhmNs8u3obNxpYxOp6QvXK9sXoYAVjUzsZx0Nm0PXJ2FnbW0Xn1REnmAwDgRABQy31f9jTQkkf97or5gGdyY0DrMOiJ/OB9zoLRG1Dd6iVWEI7yTNk0RUMTaL2AL1IxfP+OFXmgCEABOZ+vpFHeoHv5FQkZUTWSnBydJA7+bCsncl0eH41kPShmbrpZ13zOpIz8bRDgYxevZsWTUfjtSs3
```

```
1TA0DRAaO5HLHuBzTdD43vWdkhsrjYdA4pHK8Q3YMuozOZvvLpFc43dhBrnLXpPmV+wwNy
faIM9nR14hAyibcsC6Q9AZF9x2bavsy16wUGjvN1jLVg2ucIR2Cpd6sWngpzU2S1s6ZvU9
7TNpg9XTRs5PCzk40JFprOh+tKC2JsU11URTZEhliXl/IPGGQ/tmxm8tgQ+3;_WebSsoA
uth0=vPDongOyasChyToiOY0bv9+Ba5vkc5xVY1G4s1R35cd7RUxWGxg/NLoAIdgGEVye9
5Owd7K1848cpbNYrsl1u9g0zokt/4hr100CbXU316XCHe5nmwbHjiO9oN/Uz98ioH1EW8Z
Oxw4O2vuh2rXn+lzamRyvW2nBHEFM9hrZerZq2q6cEUmIsBkuGGulKc4zSViwiIcEU2y1N
qw2vdjqGAoA7elxoKThMu35ess3+Po0kLHKdfeqGc11/zymA0bXGq3jUGluTIN3R4rtQMD
jMasIXrcjDOz5irf9bmSFxqUbu5P5KGREL0cCWhIkud2slqrbWLLZzc7uCGYLBkiMg697
aKmToEuFJuJOSmWZ0E4+XLEj0y1BheMyllYBamp93izXp93LVH3Eg44JKxZaLs/S3ide7V
WbNmH887GirrEGnAGnZ2THI6roWip05XTPLTNQBQkXDovgh2xgXrnYbbNXA29K0dZ+Zwu
jls6/4LHaqfJ+bNcpumzGPC/pi9j7vK/b/x69+bsI8T
```

```
<<<
```

```
HTTP/1.1 200 OK
```

```
[Application specific content]
```

4.2 SAML 1.1 Assertion Extension

Following is a **SAML assertion** fragment that illustrates the message syntax of the SAML 1.1 Assertion Extension elements in the advice element, as specified in section [2.2.3](#).

```
<saml:Advice xmlns:ads="urn:microsoft:federation">
  <ads:WindowsIdentifiers>
    AAAAAAEAAAABAAAAAABRUAAAVU+0xvWJxlc9CDm4GAAAA9AEAAAYCAAAHAgAACAIAA
    AECAAAAAGAA
  </ads:WindowsIdentifiers>
  <ads:CookieInfoHash>
    K6GNTL15/jljype53+PFRAiOfek=
  </ads:CookieInfoHash>
  <ads:WindowsUserIdentifier>
    S-1-5-21-837636885-2507236029-1846428367-500
  </ads:WindowsUserIdentifier>
  <ads:WindowsUserName>
    ADFSVM-A\Administrator
  </ads:WindowsUserName>
</saml:Advice>
```

The raw octets of the [WindowsIdentifiers \(section 3.1.5.2.1.5\)](#) binary structure, after **base64** decoding are as follows.

```
00 00 00 00 01 00 00 00 01 04 00 00 00 00 05 15 00 00 00 15 53 ED
31 BD 62 71 95 CF 42 0E 6E 06 00 00 00 F4 01 00 00 06 02 00 00 07 02
00 00 08 02 00 00 01 02 00 00 00 02 00 00
```

The octet stream is structured as follows (see section [2.2.3.2](#)).

```
00 00 00 00  WindowsIdentifierFlags = 0
                TryLocalAccount = 0
                NoUserSid = 0
01 00 00 00  PackedSidsCount = 1 (0x00000001)
                PackedSids1
                    DomainSid
01                Revision = 1 (0x01)
    04                SubAuthorityCount = 4 (0x04)
                IdentifierAuthority[0..1] = {0, 0, ...
00 00 00 05                IdentifierAuthority[2..5] = {0, 0, 0, 5
                (0x05)}
15 00 00 00                SubAuthority1 = 21 (0x00000015)
15 53 ED 31                SubAuthority2 = 837636885 (0x31ED5315)
BD 62 71 95                SubAuthority3 = 2507236029 (0x957162BD)
```

CF 42 0E 6E	SubAuthority4 = 1846428367 (0x6E0E42CF)
06 00 00 00	RidCount = 6 (0x00000006)
F4 01 00 00	Rid1 = 500 (0x000001F4)
06 02 00 00	Rid2 = 518 (0x00000206)
07 02 00 00	Rid3 = 519 (0x00000207)
08 02 00 00	Rid4 = 520 (0x00000208)
01 02 00 00	Rid5 = 513 (0x00000201)
00 02 00 00	Rid6 = 512 (0x00000200)

The **SIDs** encoded in the structure are as follows:

- S-1-5-21-837636885-2507236029-1846428367-500
- S-1-5-21-837636885-2507236029-1846428367-518
- S-1-5-21-837636885-2507236029-1846428367-519
- S-1-5-21-837636885-2507236029-1846428367-520
- S-1-5-21-837636885-2507236029-1846428367-513
- S-1-5-21-837636885-2507236029-1846428367-512

5 Security

5.1 Security Considerations for Implementers

Security considerations for the Microsoft Web Browser Federated Sign-On Protocol Extensions are specified in the following subsections. Additionally, the security considerations outlined in [\[MS-MWBF\]](#) section 5 apply to the Microsoft Web Browser Federated Sign-On Protocol Extensions.

5.1.1 Data Integrity

Data integrity concerns, as described in [\[MS-MWBF\]](#) section 5.1.1, apply to the extensions specified in this document. Of particular concern are the **SAML advice** elements specified by the SAML 1.1 Assertion Extension. These elements are included in the **SAML assertion**, which is signed to prevent tampering (see [\[MS-MWBF\]](#) section 2.2.4.2.2).

5.1.2 Privacy

The privacy considerations in [\[MS-MWBF\]](#) section 5.1.5 apply to the extensions in this document. The extensions also introduce new privacy concerns.

The Query String Response Transfer Protocol is used to communicate **security tokens** from one party to another by using the query string of the HTTP request URL. The use of Secure Sockets Layer/Transport Layer Security (SSL/TLS) prevents the exposure of **user** information outside the services participating in the protocol (see [\[MS-MWBF\]](#) section 5.1.3); however, a GET message might provide a lesser degree of confidentiality than a POST message due to URL tracking concerns. For example, **web browser requestor** implementations might track URL history, or web proxy servers might log URLs.

The SAML 1.1 Assertion Extension provides a method for including **SIDs** in a **SAML assertion**. These SIDs might identify user identity, capabilities, or affiliations. For this reason, SIDs should not be included indiscriminately; rather, their distribution should be limited to specific **relying parties**.[<24>](#)

5.1.3 Authorization Validation and Filtering

When processing **SIDs** from an **IP/STS**, **relying parties** must ensure that the IP/STS is authorized to issue SIDs that fall under a particular set of subauthorities. This is similar to namespace collision concerns with UPN and EmailAddress **claims** (as specified in [\[MS-MWBF\]](#) section 5.1.6).[<25>](#)

5.2 Index of Security Parameters

Because the Microsoft Web Browser Federated Sign-On Protocol Extensions is an authentication protocol, the security details are in the message processing rules section.

Security parameter	Section
WindowsUserIdentifier	2.2.3.1
WindowsUserName	2.2.3.1
WindowsIdentifiers	2.2.3.1

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows Server 2003 R2 operating system
- Windows Server 2008 operating system
- Windows Server 2008 R2 operating system
- Windows Server 2012 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 1.5](#): The Windows implementation of the SAML 1.1 Assertion Extension is integrated with Windows **Active Directory**. All **SIDs** that are transmitted in the **SAML assertion** correspond to Active Directory **accounts**. In order to transmit SIDs between a **requestor IP/STS** and a **resource IP/STS** that reside in different **forests**, the requestor IP/STS forest must be a **trusted forest** of the resource IP/STS forest.

[<2> Section 1.5](#): Local configuration is used to modify several of the Windows behaviors described in this document. In all cases, the local configuration must exist before the protocol is initiated. Specific instances where local configuration is used are listed below.

By default, an **IP/STS** does not issue **security tokens** with SIDs (see section [3.1.5.2](#)). The issuance of SIDs (that is, including the WindowsIdentifier (section [3.1.5.2.1.3](#)), WindowsUserName (section [3.1.5.2.1.4](#)), and WindowsIdentifiers (section [3.1.5.2.1.5](#)) elements in issued SAML assertions) can be enabled for specific **relying parties** by using local configuration.

By default, when a **user** authenticates to a resource IP/STS by using a security token from a requestor IP/STS (see section 3.1.5.2), any SIDs in the SAML assertion are ignored (that is, the WindowsIdentifier (section 3.1.5.2.1.3), WindowsUserName (section 3.1.5.2.1.4), and WindowsIdentifiers (section 3.1.5.2.1.5) elements). Processing these SIDs (as described in section 3.1.5.2) can be enabled for specified requestor IP/STSs using local configuration. In this case, the local configuration also specifies a Windows **domain** associated with the IP/STS, so that SID filtering can be performed (as specified in section [5.1.3](#)).

When a user authenticates to a resource IP/STS by using a security token from a requestor IP/STS (as specified in section 3.1.5.2), **claims** can be mapped to SIDs based on local configuration. In this case, the local configuration also specifies the desired value of the **TryLocalAccount** flag (as specified in WindowsIdentifierFlags Structure (section [2.2.3.2.1](#))).

A special algorithm is used to determine when the Query String Response Protocol is used (as specified in section [3.2.5.1.1](#)). This algorithm uses local configuration to force the use of the protocol and for the **FileExtensionBypassList** flag.

[<3> Section 1.6](#): The Query String Response Transfer Protocol is supported only in Windows Server 2003 R2, Windows Server 2008, Windows Server 2008 R2, and Windows Server 2012.

<4> [Section 1.6](#): SAML1.1 Assertion Extension is supported only in Windows Server 2003 R2, Windows Server 2008, and Windows Server 2008 R2.

<5> [Section 1.9](#): The following URIs are used as local assignments in fields specified by the SAML 1.1 Assertion Extension: "urn:federation:activedirectory" and "urn:microsoft:federation".

<6> [Section 2.2.3.1](#): Support for this field is conditional, as specified in [CookieInfoHash Element \(section 3.1.5.2.1.2\)](#).

<7> [Section 2.2.3.1](#): WindowsUserName Element (section 3.1.5.2.1.4) specifies how Windows constructs the WindowsUserName value.

<8> [Section 3.1.1.1.1](#): The **pending result** is maintained in secure session cookies written using Set-Cookie headers (as specified in [\[RFC2965\]](#)), which are added to the HTTP 302 response to the **web browser requestor**.

<9> [Section 3.1.1.1.2](#): The recommended value is used.

<10> [Section 3.1.2](#): There are no new timers. The pending result is stored by using secure session cookies (see section [3.1.1.1.1](#)).

<11> [Section 3.1.5.2.1.1](#): The IP/STS only includes this element when it is a resource IP/STS. The value identifies Active Directory, a **Lightweight Directory Access Protocol (LDAP)** service, or a **security realm**. If the user authenticated by using Active Directory, the [ClaimSource \(section 3.1.5.2.1.1\)](#) value is "urn:federation:activedirectory". If the user authenticated by using an LDAP service, the ClaimSource (section 3.1.5.2.1.1) value is a URI associated with the LDAP service, for example "LDAP://ldap.example.com". If the user authenticated by using a SAML assertion from a Requestor IP/STS, the ClaimSource (section 3.1.5.2.1.1) value is the value of the Security Assertion Markup Language (SAML) issuer. For more information about user authentication, see [\[MS-MWBF\]](#) section 3.1.5.4.3.

<12> [Section 3.1.5.2.1.2](#): The IP/STS includes the CookieInfoHash (section 3.1.5.2.1.2) element when an authentication cookie is issued at the same time as a SAML assertion. The authentication cookie is a **base64-encoded** binary structure. Windows includes the base64-encoded SHA-1 hash value (as described in [\[FIPS180\]](#)) of the raw octets.

<13> [Section 3.1.5.2.1.3](#): By default, the IP/STS does not include the WindowsUserIdentifier (section 3.1.5.2.1.3) element. It can be enabled for specific relying parties (see section [1.5](#)). A requestor IP/STS populates this field only if the user authenticated by using Active Directory. A resource IP/STS populates this field only if the user authenticated by using a security token from a requestor IP/STS, in which case the value, if present, is copied from that security token.

<14> [Section 3.1.5.2.1.4](#): By default, the IP/STS does not include the WindowsUserName (section 3.1.5.2.1.4) element. It can be enabled for specific relying parties (see section [1.5](#)). Windows constructs the user name by including the **NetBIOS** domain name (as it is always available in this context) and Active Directory account name, separated by a backslash (for example, "DOMAIN\user name"). A requestor IP/STS populates this field only if the user authenticated by using Active Directory. A resource IP/STS populates this field only if the user authenticated by using a security token from a requestor IP/STS, in which case the value, if present, is copied from that security token.

<15> [Section 3.1.5.2.1.5](#): By default, the IP/STS does not include the WindowsIdentifiers (section 3.1.5.2.1.5) element. It can be enabled for specific relying parties (see section [1.5](#)).

A requestor IP/STS populates this field only if the user authenticated by using Active Directory. In this case, a WindowsIdentifiers (section 3.1.5.2.1.5) binary structure is constructed that contains a subset of the security identifiers (SIDs) associated with the Active Directory account. The subset consists of the user SIDs, the **global group** SIDs, and the **universal group** SIDs. That structure is base64-encoded and included in this field.

A resource IP/STS populates this field only if the user authenticated by using a security token from a requestor IP/STS. If the SAML assertion issued by the requestor IP/STS contains a **WindowsIdentifiers** field, its value is copied to this field. Otherwise, the resource IP/STS maps claims from the SAML assertion of the requestor IP/STS to SIDs based on local configuration (see section 1.5). If any claims are mapped to SIDs, those SIDs are encoded in a **WindowsIdentifiers** (section 3.1.5.2.1.5) binary structure with the **NoUserSid** flag set to 1 and the **TryLocalAccount** flag set based on local configuration (see section 1.5). The structure is base64-encoded and included in the **WindowsIdentifiers** field.

<16> [Section 3.1.6](#): There are not any new timer events. The pending result is stored by using secure session cookies (see section 3.1.1.1.1).

<17> [Section 3.2.1.1.1](#): The **aggregated result** is maintained in secure session cookies written using Set-Cookie headers (as specified in [RFC2965]), which are added to the HTTP 302 response to the web browser requestor.

<18> [Section 3.2.2](#): There are no new timers. The aggregated result is stored using secure session cookies (see section [3.2.1.1.1](#)).

<19> [Section 3.2.5.1.1](#): A special algorithm (described later) determines whether to use the Query String Response Transfer Protocol or not.

The algorithm is presented in pseudocode.

Let KnownClients be a constant set composed of the following strings, which specify **user agent** string fragments that are known to identify user agents that do not support scripting:

- "Microsoft FrontPage"
- "Microsoft Office"
- "Test for Web Form Existence"
- "Microsoft Data Access Internet Publishing Provider"
- "Microsoft-WebDAV"

Let FileExtensionBypassList be a set of strings retrieved from local configuration (see section 1.5) that identify file name extensions.

The algorithm is as follows.

```
IF local configuration forces Query String Response Transfer
Protocol
OR User-Agent is empty
OR User-Agent is in KnownClients
OR NOT User-Agent contains "Mozilla"
OR NOT Http-Verb is in (GET,POST) THEN
    Use Query String Response Transfer Protocol
ELSE
    IF local machine state indicates Windows Sharepoint Services
    AND NOT Request-URL's file extension is in
FileExtensionBypassList THEN
        Use Query String Response Transfer Protocol
    ELSE
        Use the base protocol specified in [MS-MWBF]
    END IF
END IF
```

<20> [Section 3.2.5.2](#): By default, the resource IP/STS ignores these fields and processes the security token as though the extension elements are absent. Their processing can be enabled for specific requestor IP/STSs (as specified in section 1.5). Section [3.1.5.2.1](#) specifies how the resource IP/STS

includes the fields. The **web service resource** uses the fields to provide authorization services to a web-based application on the same machine by using methods that are outside the scope of this protocol.

<21> [Section 3.2.6](#): There are no new timer events. The aggregated result is stored by using secure session cookies (as specified in section 3.1.1.1.1).

<22> [Section 3.3.1](#): Windows Internet Explorer supports the use of session and persistent HTTP cookies (for more information, see [RFC2965]). The Windows implementation of this protocol requires that web browser requestor support at least session cookies. It uses persistent cookies to preserve security realm identifiers if they are supported by the web browser requestor.

<23> [Section 3.3.5](#): The RMS 2.0 client in Windows Vista operating system with Service Pack 1 (SP1), Windows Server 2008, Windows 7 operating system, Windows Server 2008 R2, Windows 8 operating system, and Windows Server 2012 adds a *whr* parameter to the wsignin 1.0 Request Message (section [2.2.2](#)) if the wsignin 1.0 Request Message does not already contain a *whr* parameter.

<24> [Section 5.1.2](#): The inclusion of SIDs can be enabled for specific relying parties (as specified in section 1.5).

<25> [Section 5.1.3](#): A relying party that accepts security identifiers from an IP/STS in another security realm performs SID filtering based on local configuration that associates the IP/STS with a Windows domain (as specified in section 1.5). For more information about SID filtering, see [\[MS-PAC\]](#) section 4.1.2.2.

7 Change Tracking

No table of changes is available. The document is either new or has had no changes since its last release.

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